AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 2 · STUDENT LESSON

Decompose a Goal

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 1: a process should state inputs and outcomes.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Distinguish a function definition from a function call. Try the recall before opening notes.

Your learning target

I can distinguish a **function definition** from a **function call**.

I can match positional **argument**s to **parameter**s in order.

I can reuse one **function** and test that it uses the supplied values.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 2: **Decompose** a Goal

Complex tasks become manageable when divided into testable **subproblem**s.



THINKER MISSION

Break a school-morning helper into **input**s, decisions, and actions.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

A script sets capacity = 12 and used = 4. What does print(capacity - used) display?

This week's end goal

By the end, I can **decompose** a calculation into a **function** with explicit **argument**s and a returned result.

STUDENT LESSON · 2 OF 18

Notice the evidence

Investigate the supplied Decompose a Goal case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **decompose** a goal to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A fictional supply cupboard records capacity and the number of occupied spaces. The remaining-space calculation is capacity minus used.

Instead of copying the same instructions for every cupboard, define one **function** and **call** it with different values. A **function** makes a procedure reusable; its name alone does not guarantee that its instructions are correct.

Our **input** contract requires nonnegative integer counts with used no greater than capacity. The example assumes these conditions; it does not validate them. This week the **function** prints its result.

Week 9 will show how to return a value for later calculations.

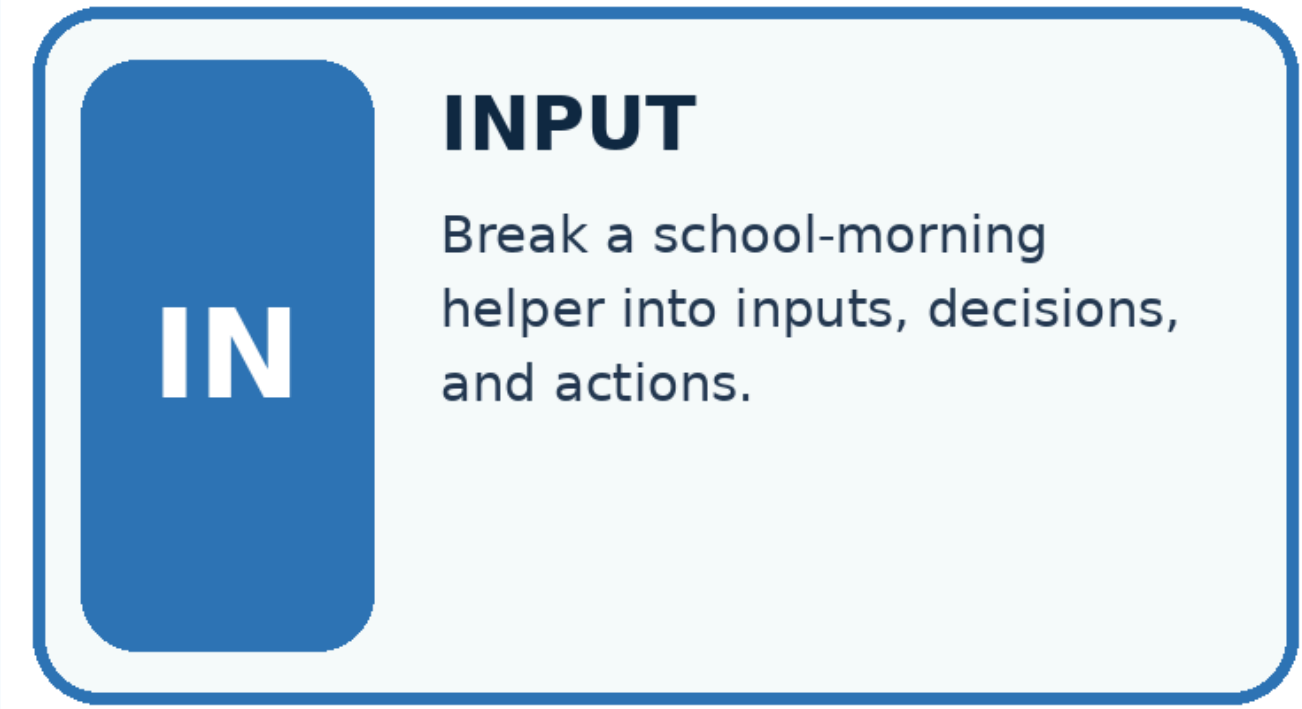
See how it works

> Follow the mechanism for Decompose a Goal. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

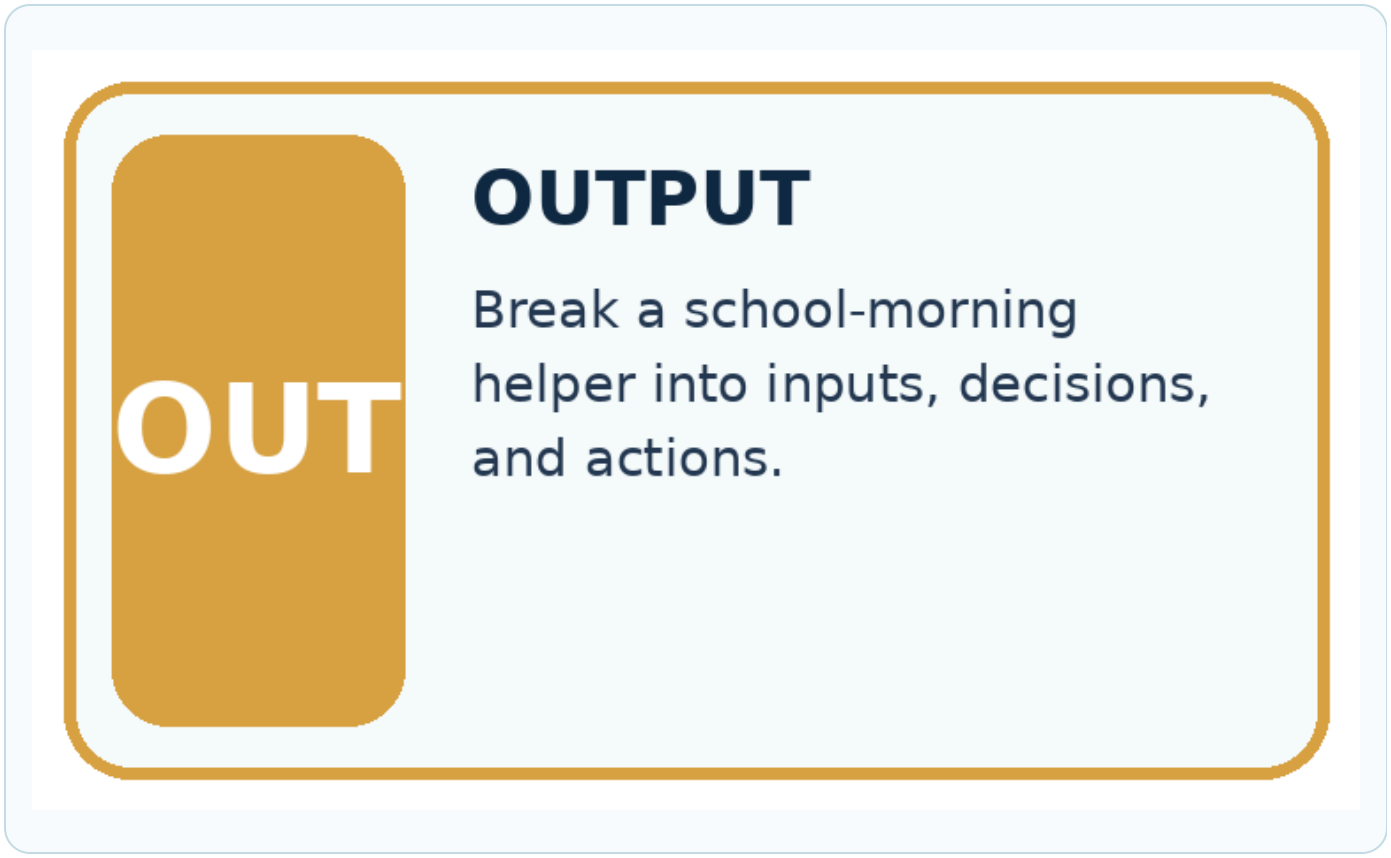
Trace the information. At each stage, ask what changed and how you could test it.



↓


↻
PROCESS
Complex tasks become manageable when divided into testable subproblems.

↓


OUT
OUTPUT
Break a school-morning helper into inputs, decisions, and actions.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
def show_remaining(capacity, used):
    remaining = capacity - used
    print(remaining)

show_remaining(12, 4)
show_remaining(10, 7)
```

Arguments matched in order

CALL	PARAMETER VALUES	PRINTED OUTPUT
show_remaining(12, 4)	capacity 12; used 4	8
show_remaining(10, 7)	capacity 10; used 7	3

Each call runs the same body with the values supplied for that call.

def introduces the **definition**. The **function** name is followed by **parameter** names in parentheses and a colon. Four-space indentation marks the body. Executing the **definition** creates the **function** but does not run its body.

The two unindented **call**s then run it, producing 8 and 3 on separate lines. A blank line helps readability; indentation determines the block.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

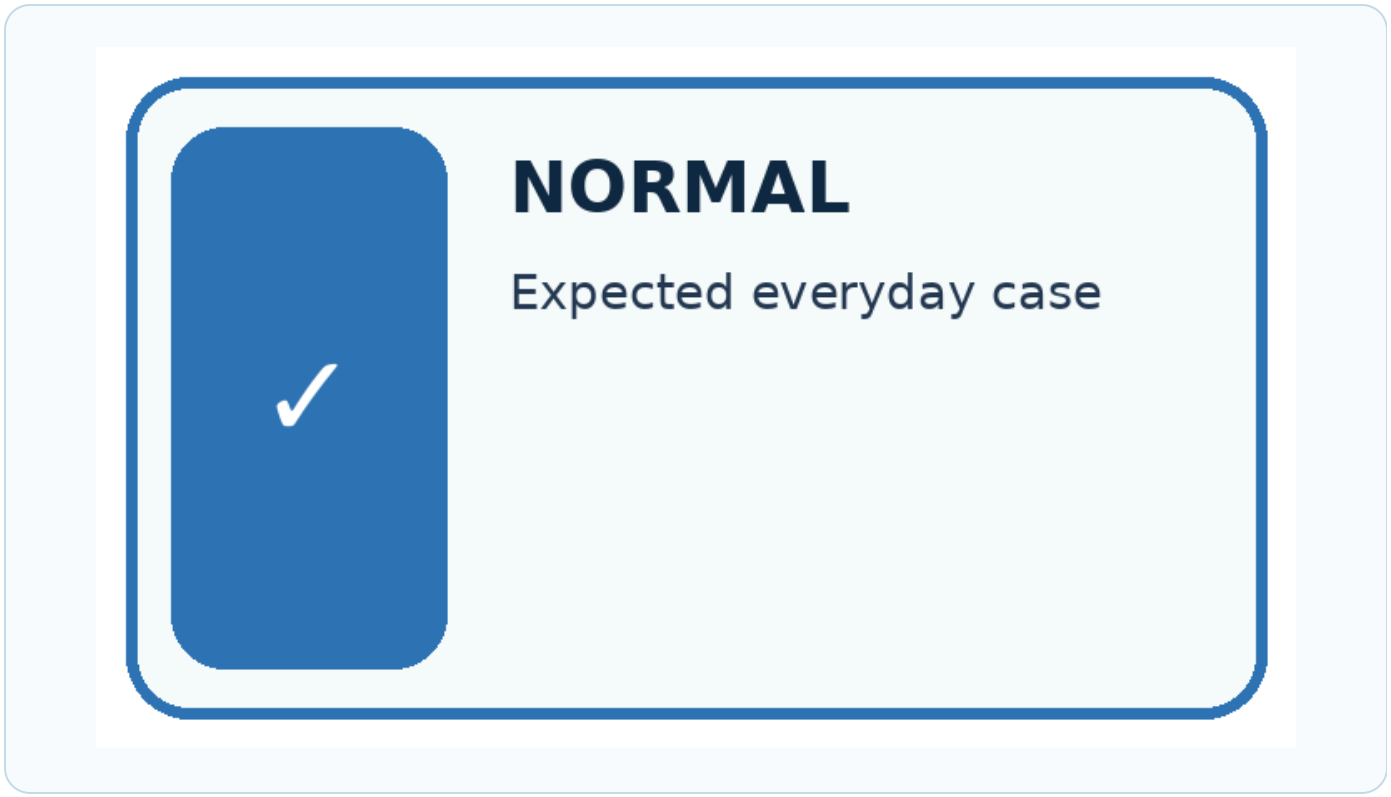
> Compare the supplied cases for Decompose a Goal. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

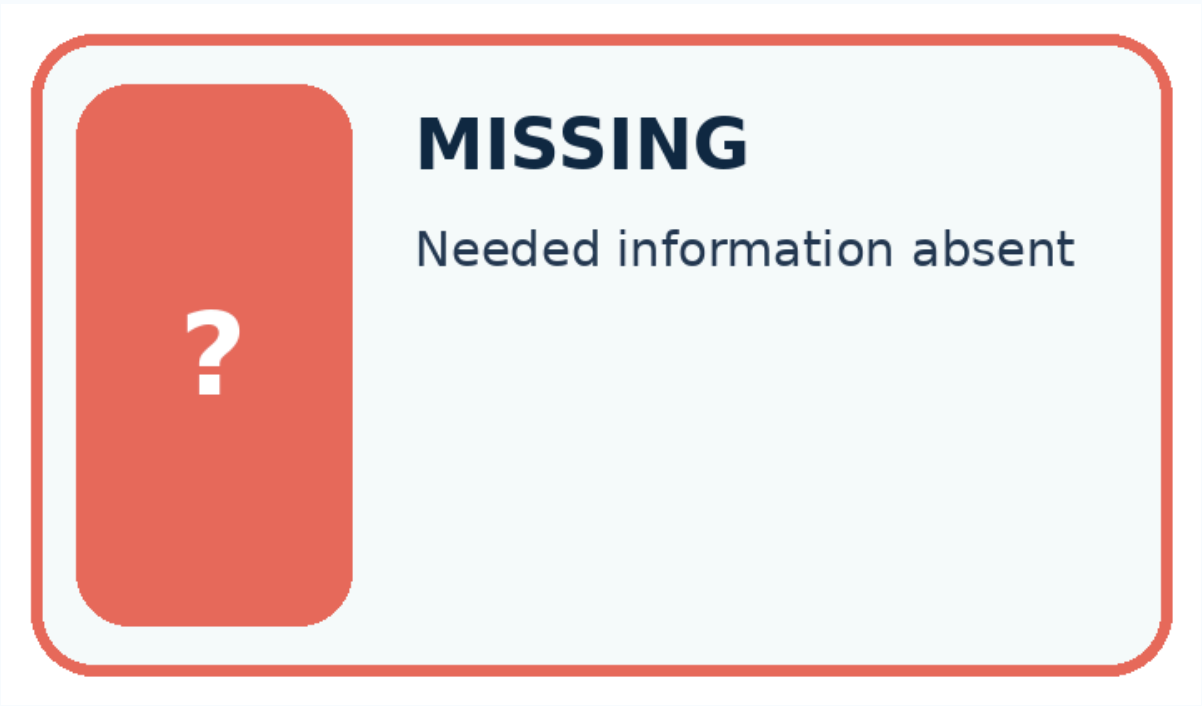
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: **Call** labels(4,0) instead.

Predicted result

Prediction: The returned result is 0, and the **call**er prints 0.

Why the result follows

Reason: Four shelves times zero labels per shelf is zero.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Decompose a Goal to explain a concrete decision, rather than only memorising definitions.

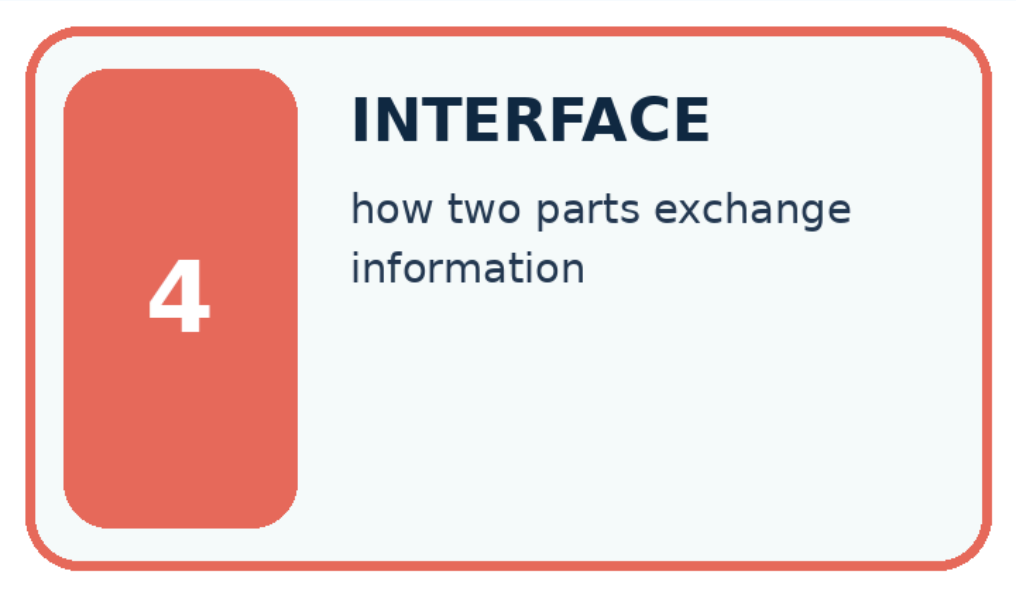
VOCABULARY LAB

Words You Can Use to Reason

A **definition** matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
DECOMPOSE
split a large task into smaller parts
2
SUBPROBLEM
one smaller solvable task





Word	My own example	Not this / boundary
decompose	Break a goal into smaller parts with clear responsibilities and connections.	
subproblem	A smaller part of a larger problem with its own inputs and intended result.	
dependency	Something a step needs from another step before it can proceed correctly.	

Word	My own example	Not this / boundary
interface	An agreed boundary specifying how two parts exchange information or actions.	

Words used in the closer explanation

Word	Meaning
Function	A named block of reusable instructions that runs when called
Definition	Code that creates a function and specifies its body
Call	An instruction that runs a function with supplied arguments
Parameter	A name in a function definition that receives an argument
Argument	A value supplied when calling a function

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Decompose a Goal, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 2 • WORKED EVIDENCE

1. Goal: help a learner leave home ready.
2. Parts: check time; check weather; check bag; choose reminder.
3. Dependency: weather must be known before the clothing suggestion.
4. Test each part separately, then test the whole flow.

1. Goal: help a learner leave home ready.

2. Parts: check time; check weather; check bag; choose reminder.

3. **Dependency**: weather must be known before the clothing suggestion.

4. Test each part separately, then test the whole flow.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Execution event	What happens
Execute the definition	Create the function; no output yet
Call with 12 and 4	Match parameters; subtract; print 8
Call with 10 and 7	Match new values; subtract; print 3
After the second call	Continue with the next statement

Worked explanation · Decompose a Goal

In show_remaining(12, 4), 12 and 4 are **argument**s; capacity and used are **parameter**s.

Positional **argument**s match from left to right. show_remaining(4, 12) produces -8, showing why order matters; it also violates our written **input** contract. Use a fresh file with the **definition** before its **call**s.

Predict the output of show_remaining(9, 9) followed by show_remaining(9, 0). Name the parameter receiving each argument. What would be printed if both calls were removed?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Decompose a Goal

A fictional library computes total spine labels needed for several equal shelves.

The facts we will use

Supplied facts

Define labels(shelves,per_shelf): return shelves*per_shelf.

Supplied facts

Call needed=labels(4,6), then print(needed). No other side effects are defined.

AI ACADEMY / AI-2/2

Decompose a Goal

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Call with arguments

The call supplies shelves=4 and per_shelf=6 locally.

2. Function body

Multiply 4 x 6 to calculate 24.



3. Return value

return sends 24 back to the caller; it does not print.



4. Caller prints

needed receives 24; print(needed) displays 24.

CHANGE ONE THING

Call labels(4,0) instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Call with arguments	The call supplies shelves=4 and per_shelf=6 locally.
2. Function body	Multiply 4×6 to calculate 24.
3. Return value	return sends 24 back to the caller; it does not print.
4. Caller prints	needed receives 24; print(needed) displays 24.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **function** separates one named job from the surrounding program. **Argument**s supply data for this **call**; return supplies its result. A print statement in the **call**er is a different action.

Clear **parameter** names help detect swapping or misusing **input**s without assuming every **function** prints automati**call**y.

What this example does not establish

Keep the stated limit

This **function** assumes compatible numeric **input**s; it does not reject negative shelves or verify real shelf contents.

Keep the stated limit

Read the labelled picture: What happens visibly if the **call**er stores needed but omits print(needed)?

Keep the stated limit

Change one thing: **Call** labels(4,0) instead. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Decompose a Goal claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Decomposition means writing every tiny action before understanding the goal. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

For each **call**, record the supplied **argument**s, expected printed **output** and observed **output**. Include different values that produce different answers, plus a valid zero-result case.

Call order controls **output** order; it does not change the subtraction rule. These **function**s use supplied data and written instructions, not a pattern learned from examples.

A learner tests only show_remaining(12, 4) and says the function works for every cupboard. Give a second valid call that exposes the fixed-number bug, state its expected output and explain what passing two tests still cannot prove.

STUDENT LESSON · 8 OF 18

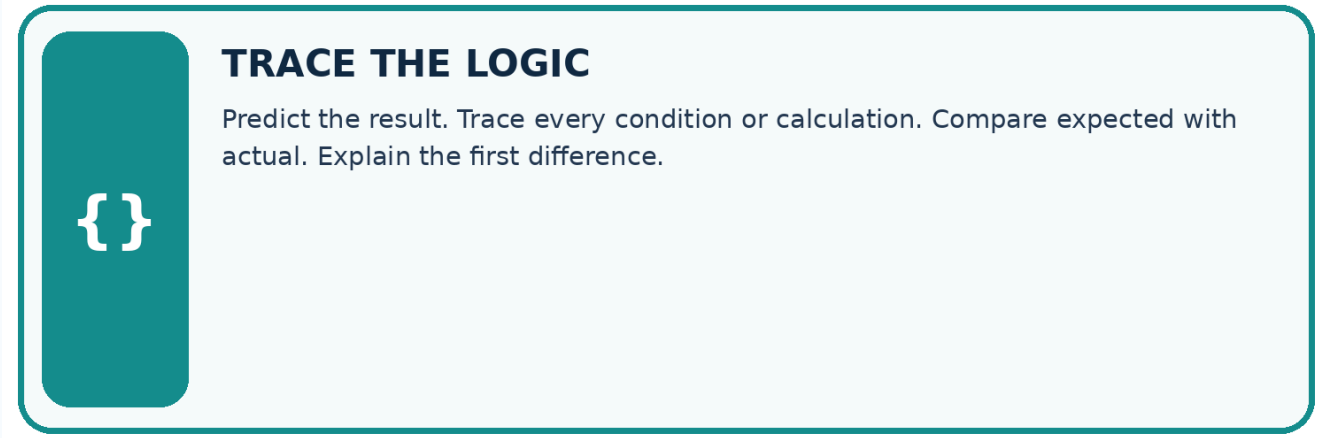
Practise together

> Practise Decompose a Goal with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

```
def show_remaining(capacity, used):
    remaining = 12 - 4
    print(remaining)

show_remaining(12, 4)
show_remaining(10, 7)
```

This faulty **function** prints 8 twice. Its fixed numbers ignore the **argument**s. Replace 12 - 4 with capacity - used, then rerun both **call**s.

Testing only the first **call** would miss the bug because its expected answer happens to be 8.

For this two-**parameter definition**, show_remaining(12) raises TypeError because an **argument** is missing. Correct the **call** to supply both values.

Naming a **function** without **call**ing it does not execute the body: the parentheses in show_remaining(12, 4) are part of the **call**.

Practise with less help: Follow the execution order

Use workbook W2 for the connected example "**Function**s and **argument**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Function**s and **argument**s. Proposed explanation: "Writing def runs the body immediately". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> What happens visibly if the **call**er stores needed but omits print(needed)?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

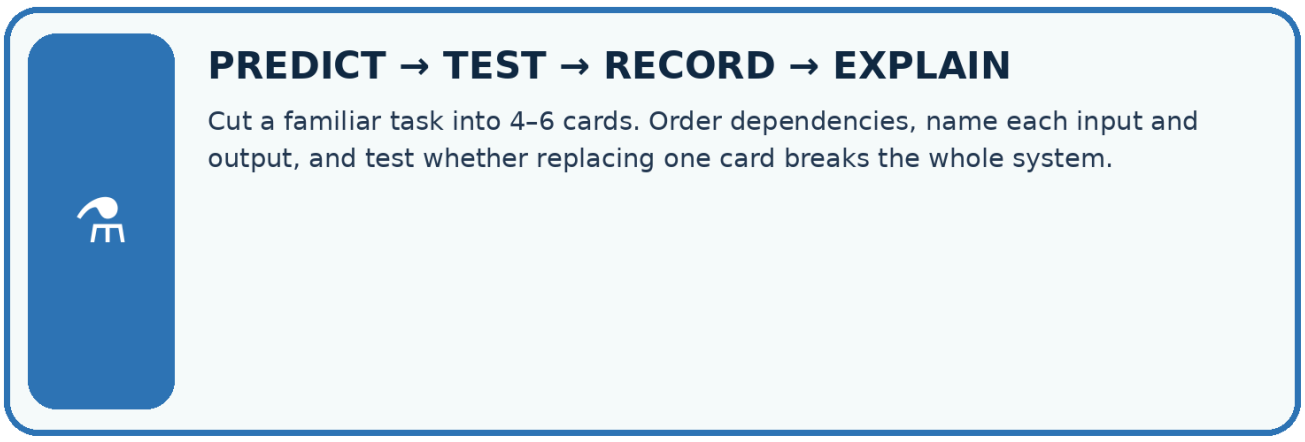
Run the investigation

> Investigate Decompose a Goal. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Cut a familiar task into 4–6 cards. Order dependencies, name each **input** and **output**, and test whether replacing one card breaks the whole system.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

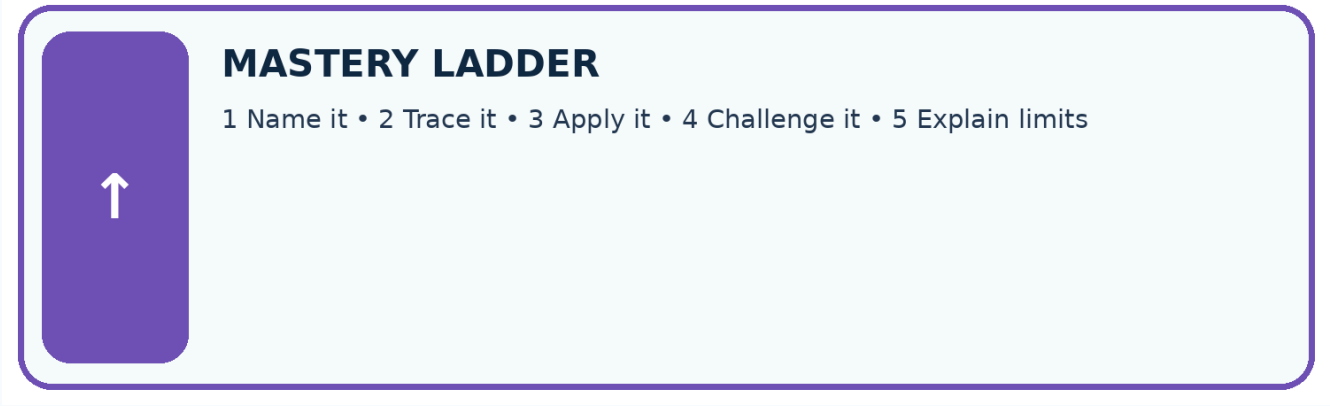
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Decompose a Goal. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

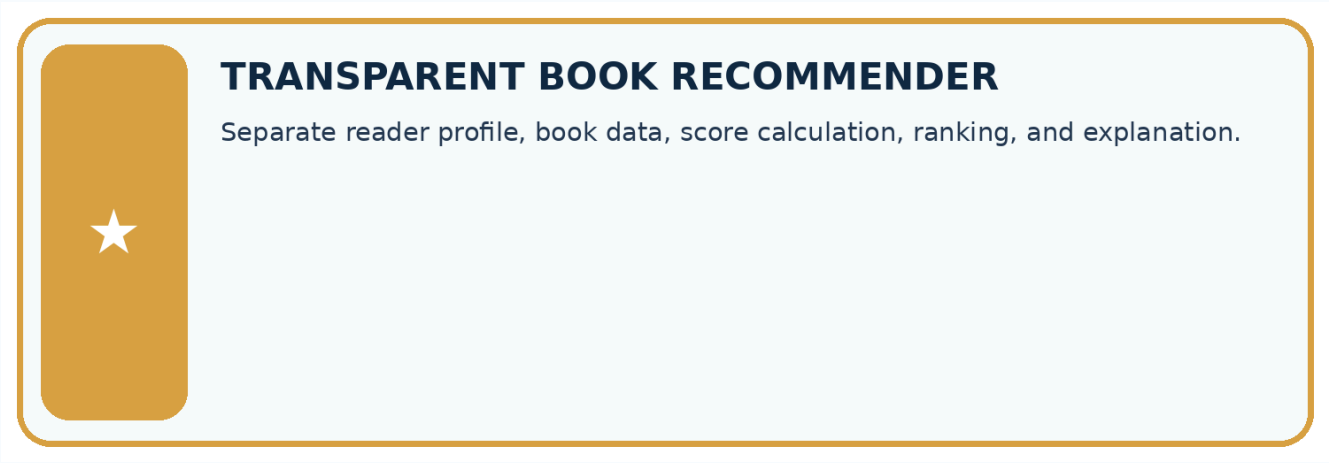
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Separate reader profile, book data, score calculation, ranking, and explanation.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Give the recommender separate validation, eligibility and scoring **function**s whose **input**s and results can be tested independently.

STUDENT LESSON · 12 OF 18

Show what you know

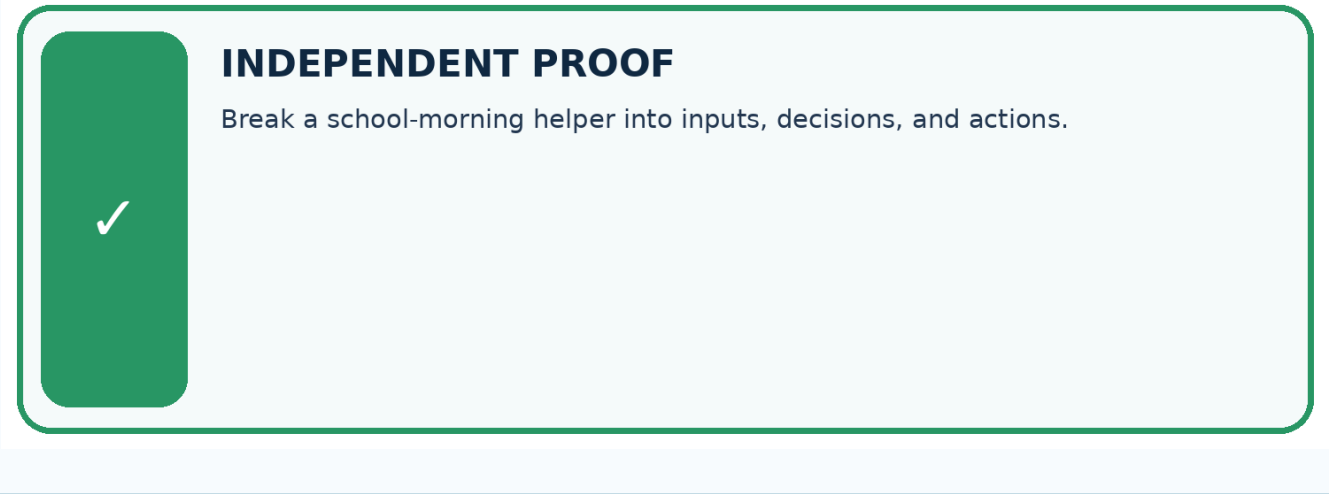
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Break a school-morning helper into **input**s, decisions, and actions.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Decompose a Goal case for supported practice. Repeated examples do not become fresh assessments.

Week 2 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A class wants to recommend one of five library books.

The question

Decompose the goal into checkable steps.

Worked explanation

Collect preferences, record book traits, score each with a declared rule, handle ties and explain the suggestion. Each step produces an inspectable record.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Decompose a recommendation into records

Evidence or stage	Value or result
Preferences	Collect reader choices
Traits	Describe five books
Scoring	Apply the declared rule
Decision	Handle ties and explain

A class wants to recommend one of five library books.

Decompose the goal into checkable steps.

Collect preferences, record book traits, score each with a declared rule, handle ties and explain the suggestion. Each step produces an inspectable record.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Create a new function independently

Each sealed box contains 6 counters. Define show_stock(boxes, loose) to print the total counters using its two **parameter**s. Assume nonnegative integer **argument**s. **Call** it with (3, 2), (0, 2) and (3, 0).

Predict each **output** and explain why show_stock(2, 3) is a different case.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 2 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A recommender must receive a topic, find matching books and show one title. Split the goal into three tasks and state what passes between them.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Separate reader profile, book data, score calculation, ranking, and explanation.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Decompose a recommender for books under 120 pages in a requested genre. A teammate scores every book before checking the page limit. Explain a clearer order and the record passed between stages.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Decompose a Goal, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional library computes total spine labels needed for several equal shelves.

By the end, I can	How I will show it
Distinguish a function definition from a function call	Explain the deciding step, show a trace or test, and state one limit.
Match positional arguments to parameters in order	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Reuse one function and test that it uses the supplied values	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Decompose a Goal without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Distinguish a **function definition** from a **function call**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional library computes total spine labels needed for several equal shelves.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? needed receives 24; print(needed) displays 24.

1. **Function** body

2. **Call**er prints

3. **Call** with **argument**s

4. Return value

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Call**er prints needed receives 24; print(needed) displays 24. A **function** separates one named job from the surrounding program.

Arguments supply data for this **call**; return supplies its result. A print statement in the **call**er is a different action.

Clear **parameter** names help detect swapping or misusing **input**s without assuming every **function** prints automati**call**y.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: **Call** labels(4,0) instead. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The returned result is 0, and the **call**er prints 0.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The returned result is 0, and the **call**er prints 0. Four shelves times zero labels per shelf is zero.

Repair a missing building block

Review Precision in Algorithms (ai-2/1). Goal: Translate a short paper calculation into a Python script. Short practice: If print(total) moves before total=total+2, what gets printed for the original **input**?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

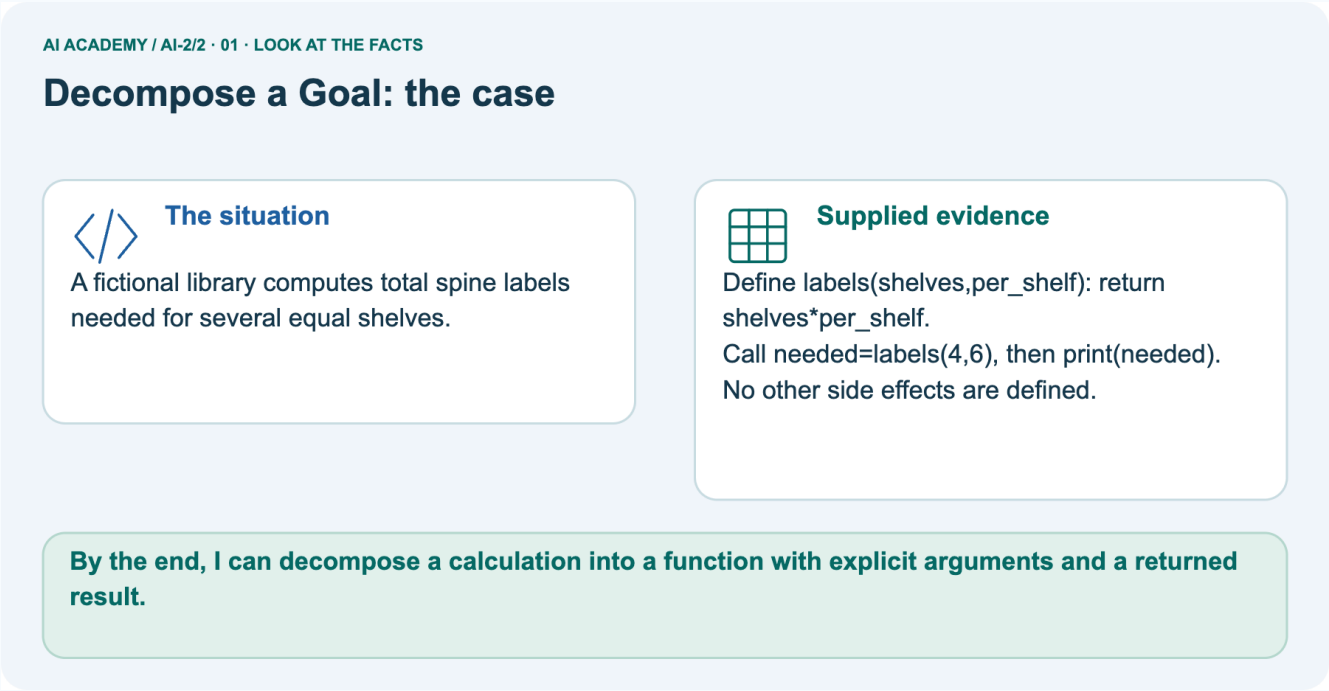
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Decompose a Goal graphic using its labels, deciding evidence and relevant limitation.

01. Decompose a Goal: the case



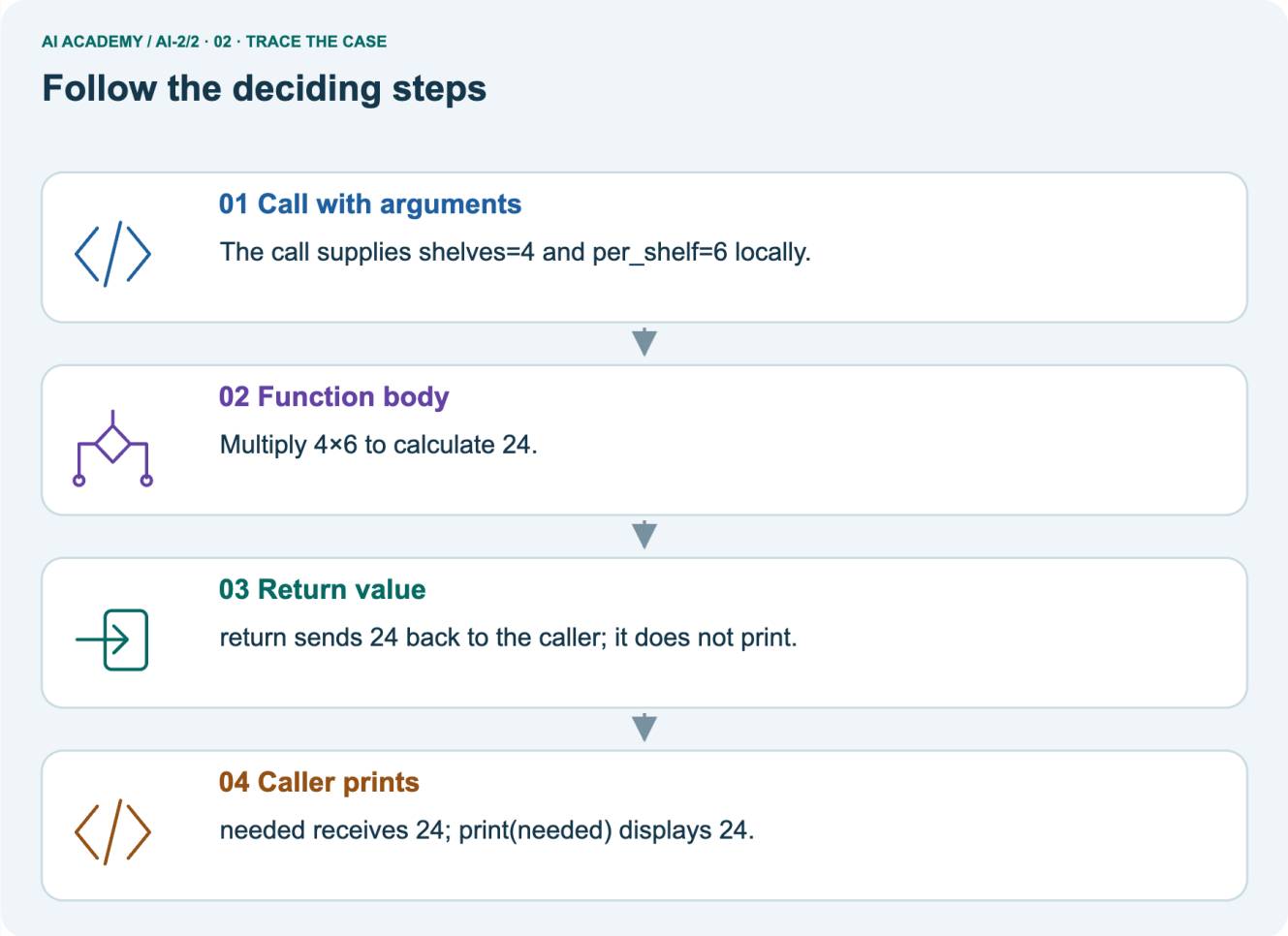
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional library computes total spine labels needed for several equal shelves. Supplied facts: Define labels(shelves,per_shelf): return shelves*per_shelf. **Call** needed=labels(4,6), then print(needed).

No other side effects are defined.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Call with **argument**s: The **call** supplies shelves=4 and per_shelf=6 lo**call**y. → **Function** body: Multiply 4×6 to calculate 24. → Return value: return sends 24 back to the **call**er; it does not print. → **Call**er prints: needed receives 24; print(needed) displays 24.

Point to each numbered stage and explain which recorded **input** it uses.

03. Arguments matched in order

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Arguments matched in order

Call	Parameter values	Printed output
show_remaining(12, 4)	capacity 12; used 4	8
show_remaining(10, 7)	capacity 10; used 7	3

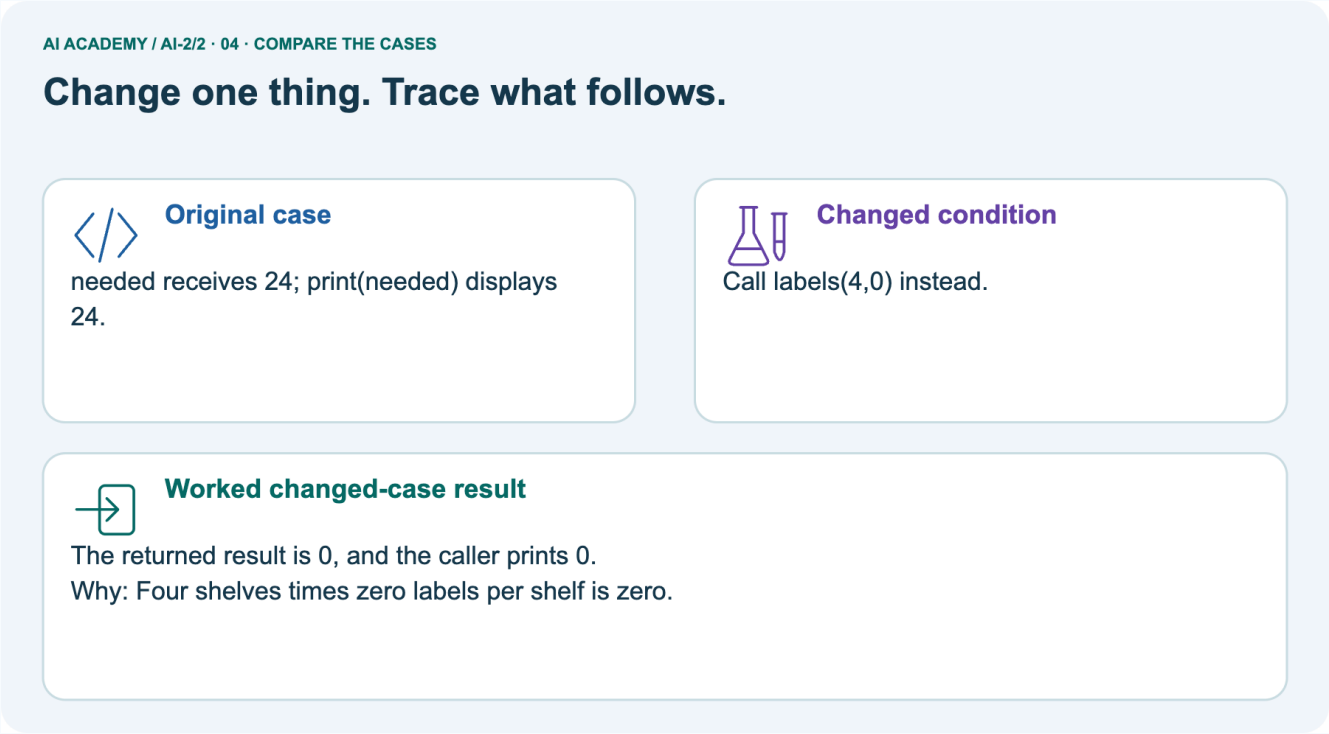
Each call runs the same body with the values supplied for that call.

Follow the labelled picture with the matching explanation. Open the vector diagram

Arguments matched in order. show_remaining(12, 4) | capacity 12; used 4 | 8. show_remaining(10, 7) | capacity 10; used 7 | 3

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



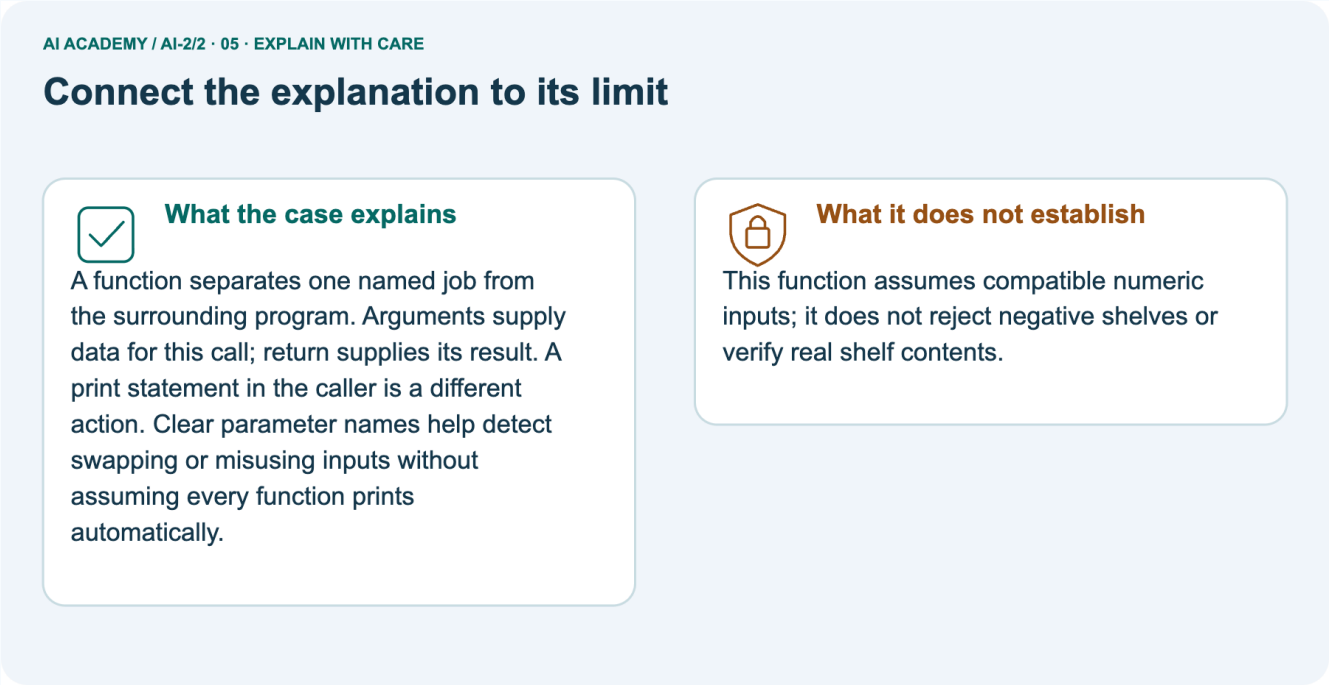
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: needed receives 24; print(needed) displays 24. Changed condition: **Call** labels(4,0) instead. Expected result: The returned result is 0, and the **call**er prints 0.

Reason: Four shelves times zero labels per shelf is zero.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

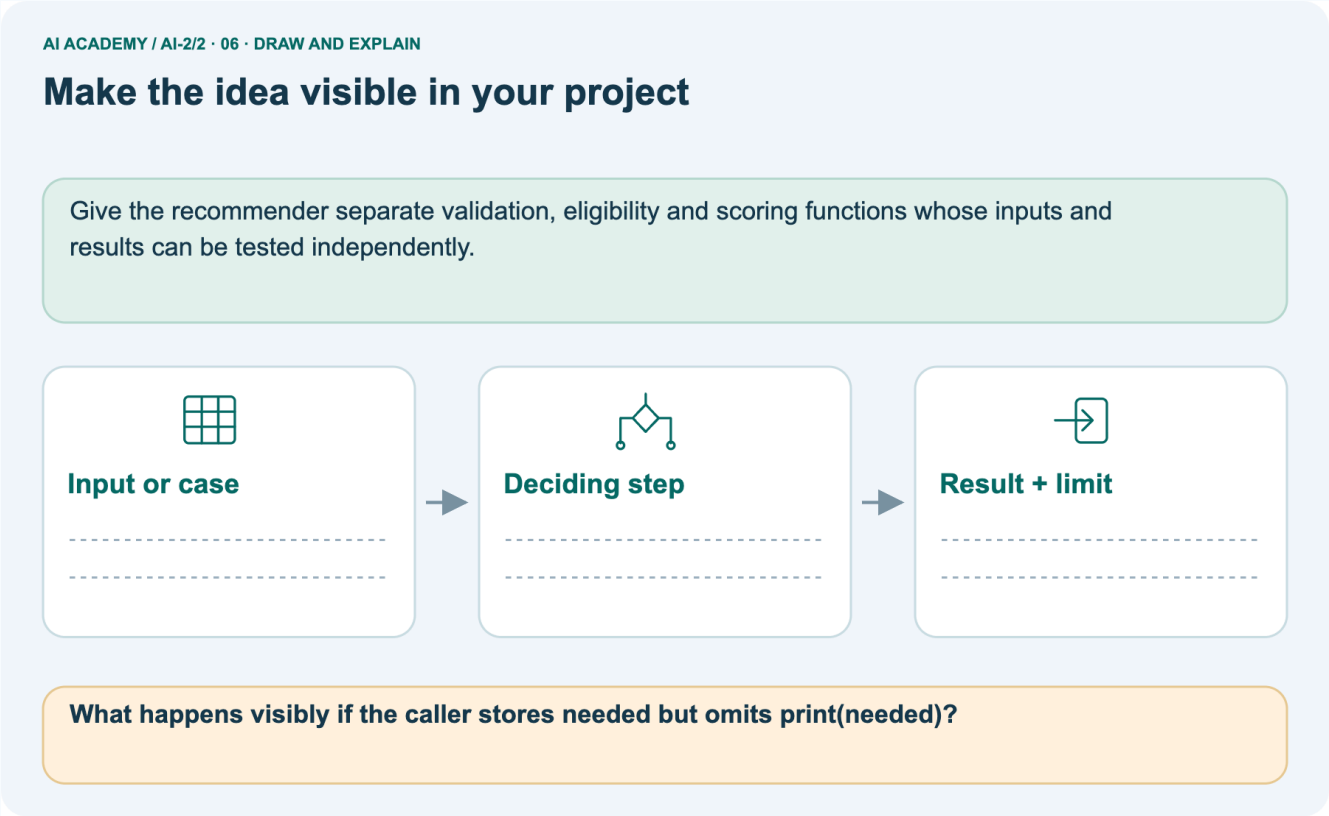
A **function** separates one named job from the surrounding program. **Argument**s supply data for this **call**; return supplies its result. A print statement in the **call**er is a different action.

Clear **parameter** names help detect swapping or misusing **input**s without assuming every **function** prints automati**call**y.

Limit: This **function** assumes compatible numeric **input**s; it does not reject negative shelves or verify real shelf contents.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Give the recommender separate validation, eligibility and scoring **function**s whose **input**s and results can be tested independently. Draw your **input**, deciding step, result and limit.

Supported question: What happens visibly if the **call**er stores needed but omits print(needed)?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.